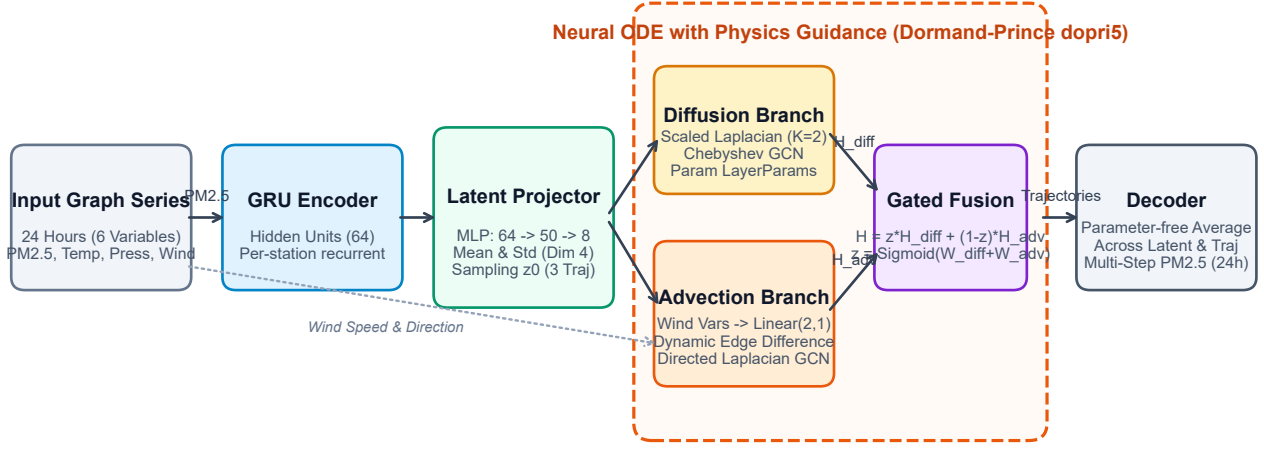

Neural ODE with Physics Guidance (Dormand-Prince dopri5)
Input Graph Series
24 Hours (6 Variables)
PM2.5, Temp, Press, Wind
PM2.5
GRU Encoder
Hidden Units (64)
Per-station recurrent
Latent Projector
MLP: 64 -> 50 -> 8
Mean & Std (Dim 4)
Sampling z0 (3 Traj)
Wind Speed & Direction
Diffusion Branch
Scaled Laplacian (K=2)
Chebyshev GCN
Param LayerParams
Advection Branch
Wind Vars -> Linear(2,1)
Dynamic Edge Difference
Directed Laplacian GCN
H_diff
H_adv
Gated Fusion
H = z*H_diff + (1-z)*H_adv
z = Sigmoid(W_diff+W_adv)
Trajectories
Decoder
Parameter-free Average
Across Latent & Traj
Multi-Step PM2.5 (24h)